



Assignment No:- 10

Q580AL

* Aim:- Data Visualization III
Download the Iris flower dataset or any other dataset into a Data frame (e.g. <https://archive.ics.uci.edu/ml/dataset/Iris>) scan the dataset and given the inference as:-

1) List down the feature and their types. (Example - numeric nominal) available in dataset.

2) create a histogram for each feature in the dataset to illustrate the feature distribution.

3) create a box plot for each feature in the dataset.

4) create a box compare distribution & identify outliers.

* Theory:-

The Iris Dataset contain features of 50 samples of three species of Iris.

(iris setosa, iris virginica and iris versicolosa)

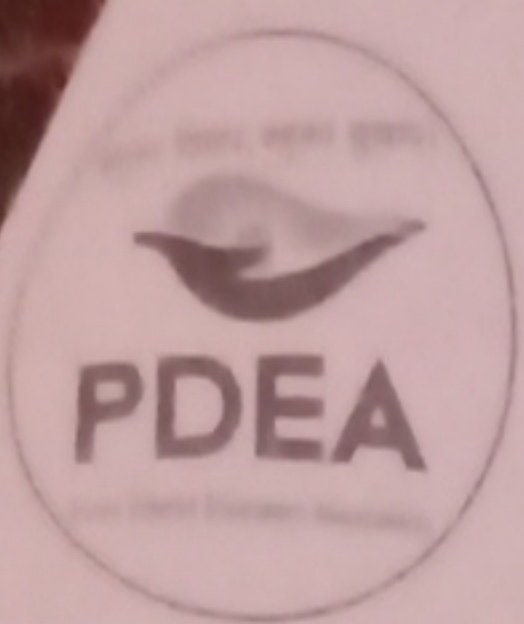
These measure were used to make or create linear discriminant model to classify the species.

The iris dataset is often used as training dataset in mlc learning mlc learning algorithms are trained to classify the different iris species based on the characteristics of the dataset. This is often used as an example of multi-class classification.

The dataset is first split into training and testing test, where 70% of the data is used for training and 30% of the for testing.

The random forest classifier is then trained on the training set and used to make prediction on the testing set, finally, the model accuracy is calculated using the accuracy - score function from sklearn library and printed out to the console.

The accuracy score represents the proportion of correctly classified instance out of all instances in the testing set.



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Manjari (Bk.), Hadapsar, Pune-412307.

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An accuracy score of 100%. Would indicate that the model has classified all instance correctly, while an accuracy score of 0%. Would indicate that the model has classified all instances correctly while on instances incorrectly.

* Why This:-

Dataset is so popular.

बहजन हिताय, बहजन सुखाय।

- The iris dataset is popular dataset for several seasons.

- First it simplest & easy-to-understand dataset. The feature of the dataset are well understood and the classification of the species is well defined.

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- This makes the dataset are well understood and the classification of the species is well defined.

- This makes the dataset easy-to-work with & understand even for beginners in data science.

- Additionally, the iris dataset is a well-established and widely used the

dataset.

- This makes it easier for beginner in data science to find useful information & example of using the data set.

* Conclusion:-

The iris dataset is a well-known & widely used dataset in data science. It is used for exploratory data analysis.